

Cloud Computing

Unit III: Virtualization in Cloud Computing

Course Code: 2301315A | TY B.Tech AI & DS | Sem VI

Topics Covered	EndSem Marks: 60 Unit Weight: ~20%
1. Introduction & Definition of Virtualization	CO3 — Bloom's Level 2 (Understand)
2. Types of Virtualization	CO3 — Bloom's Level 2 (Understand)
3. Virtualization Architecture & Software	CO3 — Bloom's Level 2 (Understand)
4. Virtual Clustering	CO3 — Bloom's Level 2 (Understand)
5. Virtualization Applications & Pitfalls	CO3 — Bloom's Level 2 (Understand)
6. Grid, Cloud & Virtualization	CO3 — Bloom's Level 2 (Understand)
7. Virtualization & Cloud Security	CO3 — Bloom's Level 2 (Understand)
8. Anatomy of Cloud Infrastructure	CO3 — Bloom's Level 2 (Understand)
9. CPU, Network & Storage Virtualization	CO3 — Bloom's Level 2 (Understand)

1. Introduction to Virtualization

1.1 Definition

Virtualization

Virtualization is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources."

In other words, Virtualization is a technique which allows sharing of a single physical instance of a resource or an application among multiple customers and organizations. It does so by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.

1.2 Key Terminologies

Virtual Machine (VM)

A virtual machine is a computer of virtual type that operates beneath a hypervisor. It provides an environment logically separated from the underlying hardware.

Hypervisor

The hypervisor is the operating system that runs on actual hardware and controls/emulates the virtual process. It is also called the Virtual Machine Manager (VMM). It is defined as Domain 0 or Dom0.

Container

Containers are lightweight virtual machines that are a subset of the same operating system instance or hypervisor. They are a collection of processes that execute with corresponding namespaces or process identifiers.

Host Machine

The physical machine on which a virtual machine is created. Manages physical resources.

Guest Machine

The virtual machine that is created on top of the host machine.

Virtual Network

A network that is logically separated and present inside servers. Such networks can be expanded across multiple servers.

Virtualization Software

Software that helps deploy virtualization on a computer device.

1.3 How Virtualization Works

The concept behind virtualization is the creation of a virtual machine over an existing operating system and hardware, known as Hardware Virtualization. In cloud computing, users share the infrastructure (not just data) with the help of virtualization.

The main usage of Virtualization Technology is to provide the applications with standard versions to cloud users. When a new version is released, the cloud provider cannot update individually due to cost. Using virtualization, all servers and software required by other cloud providers are maintained by third-party people, and cloud providers pay on a monthly or annual basis.

1.4 Benefits of Virtualization in Cloud Computing

- More flexible and efficient allocation of resources.
- Enhances development productivity.
- Lowers the cost of IT infrastructure.
- Enables remote access and rapid scalability.
- Provides high availability and disaster recovery.
- Pay-per-use of IT infrastructure on demand.
- Enables running multiple operating systems on a single physical machine.

1.5 Drawbacks (Pitfalls) of Virtualization in Cloud Computing

- The transition from an existing hardware setup to a virtualized setup requires extensive time investment — it is a time-intensive process.
- There is a lack of skilled resources to manage the transition from actual to virtual setup.
- Due to the shortage of skilled resources, implementation of virtualization calls for high-cost implementations.
- If the transition process is not handled meticulously, it also poses a security risk to sensitive data.

2. Types of Virtualization

Syllabus Note

The four primary types of virtualization are: (1) Hardware, (2) Operating System, (3) Server, and (4) Storage Virtualization. Software Virtualization is covered separately.

2.1 Hardware Virtualization

Hardware Virtualization

When the Virtual Machine Software or Virtual Machine Manager (VMM) is directly installed on the hardware system, it is known as Hardware Virtualization. The main job of the hypervisor is to control and monitor the processor, memory, and other hardware resources.

After virtualization of the hardware system, different operating systems can be installed on it and different applications can be run on those OS instances.

Usage: Hardware virtualization is mainly done for server platforms, because controlling virtual machines is much easier than controlling a physical server.

2.2 Operating System Virtualization

OS Virtualization

When the Virtual Machine Software or Virtual Machine Manager (VMM) is installed on the Host Operating System instead of directly on the hardware system, it is known as Operating System Virtualization.

Usage: OS Virtualization is mainly used for testing applications on different platforms of OS.

2.3 Server Virtualization

Server Virtualization

When the Virtual Machine Software or Virtual Machine Manager (VMM) is directly installed on the Server System, it is known as Server Virtualization.

Usage: Server virtualization is done because a single physical server can be divided into multiple servers on a demand basis and for load balancing.

2.4 Storage Virtualization

Storage Virtualization

Storage Virtualization is the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device. It is also implemented using software applications.

Usage: Storage virtualization is mainly done for backup and recovery purposes.

2.5 Comparison Table: Types of Virtualization

Type	VMM Installed On	Primary Usage	Example
Hardware	Directly on hardware	Server platforms	VMware ESXi
OS	Host Operating System	Cross-platform app testing	VirtualBox, VMware Workstation
Server	Server System	Load balancing, demand-based scaling	Microsoft Hyper-V
Storage	Via software	Backup and recovery	IBM SAN Volume Controller

2.6 Software Virtualization

Software Virtualization

Software Virtualization is a technique that allows one computer server to work with more than one virtual system. It abstracts the software installation procedure and builds virtual software installations.

Virtualized Software

A program installed inside its own self-contained unit, capable of running independently without installing on a physical machine.

The concept: Software Virtualization builds a virtual environment and allows the user to use more than one Operating System simultaneously. For example, if the user wants to use Windows and Linux at the same time, virtualization builds a virtual environment enabling both.

Types of Software Virtualization

- **OS Virtualization:** More than one Operating System works individually to complete tasks without affecting others. Each OS performs its specified job independently.
- **Application Virtualization:** Users can remotely access their applications hosted on a central server. It helps run multiple applications simultaneously by building a virtual environment.
- **Service Virtualization:** A technique to simulate the behaviors of components in the form of combination component-based applications.

Benefits of Software Virtualization

- **Time-Saving:** Helps organizations complete tasks efficiently, saving time.
- **Quick Changes:** The user can make quick changes to software according to requirements.
- **High Security:** The firewall acts as a bodyguard, preventing virus entry and keeping data secure.
- **Effective Utilization:** Available resources are used efficiently through virtual environments, enabling multiple OS usage on one computer.
- **Easy Management:** Updates are done at one location and deployed to all client systems.
- **Software Migration:** Migration from one platform to another is simplified using software virtualization.

3. Virtualization Architecture and Software

Virtualization Architecture

A virtualization architecture is a conceptual model specifying the arrangement and interrelationships of the particular components involved in delivering a virtual — rather than physical — version of something, such as an operating system (OS), a server, a storage device, or network resources.

3.1 Traditional vs. Virtual Architecture

Traditional Architecture	Virtual Architecture
One Application per OS instance.	Multiple Applications (APPs) run simultaneously.
One Operating System manages all hardware.	Multiple OS instances run on a single virtualization layer.
Direct hardware access by OS.	Virtualization Layer (Hypervisor) sits between OS and hardware.
No isolation between applications.	Strong isolation between VMs — failure of one does not affect others.
Hardware underutilization is common.	Maximizes hardware utilization through multiple VMs.

3.2 Hypervisor-Based Virtualization

Virtualization is commonly hypervisor-based. The hypervisor isolates operating systems and applications from the underlying computer hardware so the host machine can run multiple virtual machines (VMs) as guests that share the system's physical compute resources, such as processor cycles, memory space, and network bandwidth.

Type 1 Hypervisor (Bare-Metal)

Type 1 Hypervisor

Type 1 hypervisors, also called bare-metal hypervisors, run directly on top of the host system hardware. They offer high availability and resource management. Their direct access to system hardware enables better performance, scalability, and stability.

Examples of Type 1 hypervisors: Microsoft Hyper-V, Citrix XenServer, VMware ESXi.

Type 2 Hypervisor (Hosted)

Type 2 Hypervisor

A Type 2 hypervisor, also known as a hosted hypervisor, is installed on top of the host operating system, rather than sitting directly on hardware. Each guest OS or VM runs above the hypervisor. The convenience of a known host OS eases system configuration and management. However,

the addition of a host OS layer can limit performance and expose possible OS security flaws.

Examples of Type 2 hypervisors: VMware Workstation, Virtual PC, Oracle VM VirtualBox.

3.3 Comparison: Type 1 vs. Type 2 Hypervisor

Type 1 (Bare-Metal)	Type 2 (Hosted)
Runs directly on hardware.	Runs on top of a host operating system.
Better performance and scalability.	Slightly lower performance due to OS overhead.
Higher security — no host OS vulnerabilities.	May expose host OS security flaws.
Used in enterprise/data center environments.	Used for development/testing on personal machines.
Examples: VMware ESXi, Hyper-V, XenServer.	Examples: VirtualBox, VMware Workstation.

3.4 Containerization (Alternative to Hypervisor)

Containerization

The main alternative to hypervisor-based virtualization is containerization. OS virtualization, for example, is a container-based kernel virtualization method. In this architecture, an OS is adapted so it functions as multiple discrete systems, making it possible to deploy and run distributed applications without launching an entire VM for each one. Multiple isolated systems, called containers, run on a single control host and all access a single kernel.

Virtual Machines (Hypervisor)	Containers
Each VM includes a full OS.	Containers share the host OS kernel.
Heavyweight — more resource usage.	Lightweight — faster startup.
Strong isolation between VMs.	Process-level isolation.
Used for running diverse OS environments.	Used for microservices and app deployment.
Example: VMware ESXi.	Example: Docker, Kubernetes.

4. Virtual Clustering in Cloud Computing

Virtual Cluster

A virtual cluster is a many-to-one virtualization technology which can form a routing system from multiple common devices connected through a switching network, while performing the same as a single logical router to all external appearances.

Virtual clusters are built with VMs installed at distributed servers from one or more physical clusters. The VMs in a virtual cluster are interconnected logically by a virtual network across several physical networks.

4.1 Properties of Virtual Clusters

- The virtual cluster nodes can be either physical or virtual machines. Multiple VMs running with different OSES can be deployed on the same physical node.
- A VM runs with a guest OS, which is often different from the host OS that manages the resources in the physical machine where the VM is implemented.
- The purpose of using VMs is to consolidate multiple functionalities on the same server. This greatly enhances server utilization and application flexibility.
- VMs can be replicated (colonized) in multiple servers to promote distributed parallelism, fault tolerance, and disaster recovery.
- The size (number of nodes) of a virtual cluster can grow or shrink dynamically, similar to the way an overlay network varies in size in a peer-to-peer (P2P) network.
- The failure of any physical nodes may disable some VMs installed on the failing nodes. But the failure of VMs will NOT pull down the host system.

Exam Important

Key distinction: VM failure $\neq$ host system failure. Physical node failure = VM failure (on that node only). This is a favourite exam point.

5. Virtualization Application and Pitfalls

5.1 Application Virtualization

Application Virtualization

Application virtualization is technology that allows users to access and use an application from a separate computer than the one on which the application is installed. IT admins can set up remote applications on a server and deliver the apps to an end user's computer. For the user, the experience of the virtualized app is the same as using an installed app on a physical machine.

How Application Virtualization Works

The most common approach is the server-based approach. The IT administrator implements remote applications on a server inside an organization's datacenter or via a hosting service. Application virtualization software delivers the applications to a user's desktop or connected device. The user accesses and uses the application as though it were locally installed, and user actions are conveyed back to the server for execution.

Benefits of Application Virtualization

- Simplified Management: IT admins manage applications centrally rather than on each individual machine.
- Scalability: Applications can be rapidly scaled out to more users without re-installation.
- Security: Centralized control means better access management and security enforcement.

5.2 Pitfalls of Virtualization

Definition

Pitfalls of Virtualization are the challenges, risks, and failures that organizations face when adopting virtualization, despite its benefits.

- Capacity and costs fail to meet expectations.
- Application fails to run (or fails to run adequately) in a virtual environment.
- Licensing cost increase and some software is not supported in virtualized environments.
- Managers, executives, users, and other 'server huggers' resist virtualization.
- Virtualization exposes unforeseen security risks.

6. Grid, Cloud and Virtualization

6.1 Virtualization in Grid Computing

Grid Computing

Grid computing is a distributed computing model where the primary focus lies in secure resource sharing — access to computers, software, and data in a dynamic environment. Sharing of those resources has to be fine-grained and highly controlled.

Foster's Three-Point Checklist for Grid

- Delivery of nontrivial qualities of service.
- Usage of standard, open, general-purpose protocols and interfaces (e.g., for inter-communication).
- Coordination of resources that are not subject to centralized control.

OS virtualization in a grid environment provides the benefit of running multiple applications requiring different OS instances on a single computer, maximizing resource utilization across distributed systems.

Grid computing enables the virtualization of distributed computing resources such as processing, network bandwidth, and storage capacity to create a single system image, granting users and applications seamless access to vast IT capabilities.

6.2 Virtualization vs. Grid Computing

Virtualization	Grid Computing
One physical machine simulates multiple virtual machines.	Many machines in a network form a single logical system.
Single ownership of the physical machine.	Multiple people/organizations can own nodes.
Improves security for cloud environments.	More economical for compute-intensive tasks.
Work runs within a single virtualized host.	Work is split and distributed over the network.
Focus: resource isolation and simulation.	Focus: resource sharing and distributed processing.

6.3 Virtualization in Cloud Computing

In cloud computing, virtualization allows users to share infrastructure rather than only data. The key role is enabling multiple tenants to use the same physical hardware securely and efficiently. Cloud providers use virtualization to maintain standard software versions and update them centrally, eliminating the need to update each client individually.

6.4 Virtualization and Cloud Security

Virtualized Security

Virtualized security, or security virtualization, refers to security solutions that are software-based and designed to work within a virtualized IT environment. This differs from traditional, hardware-based network security, which is static and runs on devices such as traditional firewalls, routers, and switches.

Virtualized security is flexible and dynamic. Instead of being tied to a device, it can be deployed anywhere in the network and is often cloud-based. This is key for virtualized networks where operators spin up workloads and applications dynamically — virtualized security allows security services and functions to move around with those dynamically created workloads.

Benefits of Virtualized Security

- **Cost-effectiveness:** Maintains a secure network without large spending on expensive proprietary hardware. Pricing is often usage-based.
- **Flexibility:** Virtualized security functions can follow workloads anywhere — across multiple data centers, multi-cloud, and hybrid cloud environments.
- **Operational Efficiency:** Quicker and easier to deploy than hardware-based security. Security tasks can be automated, freeing up IT teams.
- **Regulatory Compliance:** Traditional hardware-based security cannot keep up with the demands of a virtualized network; virtualized security maintains compliance.

How Virtualized Security Works

Virtualized security takes the functions of traditional security hardware appliances (firewalls, antivirus) and deploys them via software. Additional functions include:

- Inserting security controls (such as encryption) between the application layer and the underlying infrastructure.
- Micro-segmentation to reduce the potential attack surface.
- Can be implemented directly on a bare-metal hypervisor or as a hosted service on a VM.

Types of Virtualized Security

- **Segmentation:** Making specific resources available only to specific applications and users by controlling traffic between different network segments or tiers.
- **Micro-segmentation:** Applying specific security policies at the workload level to create granular secure zones. Divides a data center into segments and allows IT teams to define security controls for each segment individually.
- **Isolation:** Separating independent workloads and applications on the same network. Particularly important in a multi-tenant public cloud environment.

Physical Security vs. Virtualized Security

Physical (Traditional) Security	Virtualized Security
Hardware-based, inflexible, static.	Software-based, flexible, dynamic.
Tied to specific devices at fixed network points.	Can be deployed anywhere in the network.

Physical (Traditional) Security	Virtualized Security
Focused on protecting the network perimeter.	Moves with workloads across the entire environment.
Relies on port and protocol filtering (ineffective in virtualized environments).	Adapts to dynamically assigned addresses and ports.
Cannot secure hybrid/multi-cloud environments adequately.	Ideal for hybrid and multi-cloud environments.

7. Cloud Security

Cloud Security

Cloud security, also known as cloud computing security, is a collection of security measures designed to protect cloud-based infrastructure, applications, and data. These measures ensure user and device authentication, data and resource access control, and data privacy protection. They also support regulatory data compliance.

Cloud security is employed to protect a company's data from Distributed Denial of Service (DDoS) attacks, malware, hackers, and unauthorized user access or use.

7.1 Planning of Cloud Security

As security is a major concern in cloud implementation, an organization must plan for security based on the following three main factors:

- Resources that can be moved to the cloud are picked and their sensitivity risk is tested.
- The type of cloud (public, private, hybrid) is to be considered.
- The risk in deployment of the cloud depends on the types of cloud and service models.

7.2 Types of Cloud Computing Security Controls

Key Point

There are 4 types of cloud computing security controls: Deterrent, Preventive, Detective, and Corrective. Remember: DPDC.

Control Type	Function	Purpose
Deterrent Controls	Designed to block nefarious attacks on a cloud system.	Helpful when there are insider attackers.
Preventive Controls	Make the system resilient to attacks by eliminating vulnerabilities.	Prevent attacks from succeeding.
Detective Controls	Identifies and reacts to security threats.	Examples: Intrusion detection software, network security monitoring tools.
Corrective Controls	Activated in the event of a security attack to limit damage.	Minimize impact of security breaches.

7.3 Importance of Cloud Security

- **Centralized Security:** Cloud security centralizes protection. Managing all devices and endpoints centrally enhances traffic analysis and web filtering, resulting in fewer policy and software updates.
- **Reduced Costs:** Investing in cloud security reduces expenditure on hardware and manpower in administration.

- **Reduced Administration:** Eliminates manual security configuration and constant security updates.
- **Reliability:** Cloud security systems are highly reliable and can be accessed from anywhere with any device, with proper authorization.

8. Anatomy of Cloud Infrastructure

Cloud computing is continuously changing to meet customer demands in terms of software and hardware. Operations involving hardware and data are the main players and they are not cost-effective. Cloud's on-demand infrastructure makes them cheaper and more efficient.

8.1 Cloud Stack

Cloud Stack

The infrastructure stack for delivering web applications by cloud computing providers. It is a managed cloud stack consisting of two main parts: (1) Cache, Dynamic Code, Attached Storage, and Data Logic; and (2) Stable and Efficient OS, Security Features, and Business Logic in a Programming Language.

Cloud computing environment separates the computing environment away from the developers and helps them focus on improving their applications.

Cloud Stack Key Details

- Every cloud platform includes a virtual machine language and a gateway for web services. Language functions are closely linked with the parent OS and their native libraries are removed.
- Cloud services always bundle language runtime dynamically for efficient interpretation across many application instances.
- Dynamic applications reside in application state and logic through database and file storage, placed inside cloud services in an isolated and specialized layer.
- Static files are categorized based on size: files < 1 MB can be consumed in a single request; files > 1 MB need to be split into parts for sequenced download.

8.2 Cloud Consumers

Web Application Developers form the primary consumer base. New small web applications may experience exponential growth and use cloud computing stacks for faster performance. Enterprise applications deploy different cloud models:

- SMBs and large IT companies are replacing in-house maintenance with IaaS (Infrastructure as a Service).
- Common functions like project management, employee tracking, and payroll are deployed as SaaS.
- Platforms like Windows Azure, Salesforce Force.com, and Google App Engine have strong support for back-office add-ons.

8.3 Virtual Infrastructure

Virtual Infrastructure

Virtual infrastructure is the concept where the internet evolves from a shared communication network to a pool of interconnected resources that

can be shared. Virtualization abstracts services and physical resources, simplifying management and offering great flexibility in resource usage.

Properties of the Virtual Machine in Virtual Infrastructure

- Provides an environment where non-trusted applications can be run.
- Adopts isolation techniques.
- Allows dynamic deployment of applications (portability).
- Applies OS optimization.
- Manages resources as a single service.

9. CPU Virtualization, Network and Storage Virtualization

9.1 CPU Virtualization

CPU Virtualization

CPU Virtualization is the process of making a CPU function similarly to that of two separate CPUs. It allows users to run different operating systems simultaneously on a single physical machine.

Key Points of CPU Virtualization

- Privileged instructions run only in privileged mode.
- Control-sensitive instructions that tend to change memory mappings or communicate with other devices are managed.
- Behavior-sensitive instructions that tend to change resource configuration are controlled.
- By adopting CPU virtualization, two separate CPUs resemble a single CPU — two systems running in a single system.
- For example, an Apple Mac can be virtualized to run Windows as well.

CPU Virtualization ≠ Multitasking ≠ Multi-threading

Multitasking = running multiple applications at a time. Multi-threading = more than one CPU running applications carrying out two actions simultaneously. CPU Virtualization = making one CPU appear as two separate CPUs, each running its own full OS.

9.2 Network Virtualization

Network Virtualization

Network Virtualization facilitates the running of multiple networks and multiple consumers over a shared substrate. It is a method that combines available resources by splitting bandwidth into channels and assigning them to devices or users in real time. Each channel is independently secured and every consumer has shared access to all resources on the network.

Advantages of Network Virtualization

- Managing the network becomes easier and less time-consuming for network administrators.
- Productivity and efficiency are improved.
- Files, images, programs, and folders can be managed centrally.
- Storage media such as hard drives and tape drives can be added, removed, and shared easily.

Types of Network Virtualization

- External Format: Multiple local networks are combined or subdivided into virtual networks to improve efficiency. VLAN and network switch are components of this format.

- Internal Format: A single system is configured with containers or hypervisors such as Xen/KVM domain to control VNICs. Improves overall efficiency since applications are isolated.

Components of a Virtual Network

- Network hardware components: network switch, adapters (NIC).
- Network elements: firewalls.
- VLANs and VMs.
- Network storage devices.
- Network mobile elements: tablets, mobiles.
- Network media: Ethernet cards and fibre channels.

9.3 Storage Virtualization

Storage Virtualization

Storage Virtualization is a concept where storage devices are virtualized. As a result, better functionality, proper maintenance of storage devices, and efficient backup procedures can be achieved. A storage system is also called a storage array or disk array. It is complex and controlled by special systems providing high data protection.

Types of Storage Virtualization

- Block Virtualization: Separates logical storage from physical storage. Accessing can be done without knowledge of where the physical storage is located and its nature (heterogeneous). Provides greater flexibility in managing storage devices and users.
- File Virtualization: Takes care of NAS (Network Attached Storage) by eliminating the dependencies between file level and file location. Enables optimization of storage and easier migration of storage devices.

Benefits of Storage Virtualization

- Non-disruptive data migration: Ability to migrate data without disturbing concurrent I/O access.
- Improved utilization: Utilization is increased by pooling and migration. When all storage media are pooled, the administrator can easily maintain devices and assign disks to users.

Block Virtualization	File Virtualization
Separates logical and physical storage.	Eliminates dependencies between file level and location.
Access without knowing physical location.	Facilitates optimization and migration of storage.
Works with heterogeneous storage.	Works with NAS environments.
Greater flexibility for storage admins.	Easier storage migration and management.

10. Probable Exam Questions (Unit III)

Exam Strategy

Unit III carries approx. 20% of EndSem marks (60 marks total). Focus on: definitions (exact), comparisons (tables), and 4-6 mark descriptive answers.

2-Mark Questions (Definition Type)

1. Define Virtualization in Cloud Computing.
2. What is a Hypervisor? Differentiate Type 1 and Type 2.
3. Define Virtual Machine and Container.
4. What is Hardware Virtualization?
5. Define Storage Virtualization.
6. What is Virtual Clustering?
7. What is CPU Virtualization?
8. Define Virtualized Security.

4-6 Mark Questions (Descriptive)

9. Explain the types of Virtualization with diagrams/examples.
10. Compare Type 1 and Type 2 Hypervisors. Give examples.
11. Explain Virtualization Architecture and Software.
12. What are the benefits and drawbacks of Virtualization in Cloud Computing?
13. Explain Virtual Clustering. What are its properties?
14. Explain the concept of Virtualized Security. How does it differ from physical security?
15. Explain Cloud Stack as part of the Anatomy of Cloud Infrastructure.
16. Explain Network Virtualization. What are its types and components?
17. Explain Storage Virtualization. Differentiate Block and File Virtualization.
18. Explain the pitfalls of Virtualization.
19. What are the four types of Cloud Security Controls? Explain each.
20. Compare Grid Computing and Virtualization.

Quick Revision Glossary

Term	One-Line Answer
Virtualization	Creating a virtual version of hardware/OS/network/storage.
Hypervisor / VMM	Software layer that creates and manages VMs.
Type 1 Hypervisor	Runs directly on hardware. Ex: VMware ESXi, Hyper-V.
Type 2 Hypervisor	Runs on host OS. Ex: VirtualBox, VMware Workstation.
Container	Lightweight VM sharing host OS kernel. Ex: Docker.
Hardware Virtualization	VMM installed on hardware to run multiple OS.
OS Virtualization	VMM on host OS for cross-platform testing.

Term	One-Line Answer
Server Virtualization	One physical server divided into multiple virtual servers.
Storage Virtualization	Multiple storage devices presented as one logical device.
Block Virtualization	Separates logical and physical storage (heterogeneous access).
File Virtualization	Eliminates NAS file-location dependencies.
Virtual Cluster	VMs on distributed servers forming a single logical cluster.
App Virtualization	Remote app access from central server as if locally installed.
CPU Virtualization	One CPU appears as two; runs multiple OSes simultaneously.
Network Virtualization	Bandwidth split into channels assigned to users/devices.
VLAN	Virtual Local Area Network — logical network segmentation.
Virtualized Security	Software-based security in virtualized environments.
Micro-segmentation	Granular security zones at workload level.
Cloud Stack	Infrastructure layers for delivering cloud web applications.
Grid Computing	Distributed computing — many systems, decentralized control.